

ERICA (KELA) LIU

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EDUCATION

Columbia University

MS in Computer Science

Expected Dec 2026

GPA: 3.99 / 4.33

University of California, Santa Barbara

BS in Computer Science, Minor in Philosophy

Jun 2025

Dean's Honor | Honors Distinction in Major | GPA: 3.92 / 4.00

EXPERIENCE

Personal Project | AI Engineer

Aug 2026 – Sep 2026

Indie-Help: LLM-Powered RAG System for Game Development

- Designed a hybrid retrieval system with dense embeddings + BM25, fused via Reciprocal Rank Fusion, Cohere cross-encoder reranking, and a dedicated verification stage that hard-rejects any LLM-claimed citation not verbatim-present in its source chunk, eliminating hallucinated citations architecturally.
- Architected a RAG pipeline in LangGraph with a self-correcting conditional retry edge, automatically expanding its search when zero grounded evidence is retrieved before falling back to abstention.
- Built and debugged an automated evaluation harness against a 27-question golden set engineered around six specific failure modes. Benchmarked against Ragas' default metrics and a custom rubric, the latter rising from 5.43 (a simple single-call RAG baseline) to 8.14 (refined pipeline).

Columbia University | Research Assistant

Jan 2026 – Aug 2026

Creatures in TV: Full-Stack Generative AI & Computer Vision Application

- Architected and deployed a full-stack multimodal web app on AWS ECS Express mode, orchestrating generative AI and advanced computer vision to dynamically composite sprite-sheet animations into spatially-aware environments.
- Implemented custom depth-sorting logic leveraging SAM2 spatial masks to dynamically assign precise Z-index layers, maintaining consistent and accurate occlusion whether 2D creatures pass in front of or behind real-world objects in the image.
- Designed and implemented a highly intuitive frontend interface that bridges complex CV backend pipelines, empowering users to interact with creature animations within their personal photos in real-time.

Linx Robot | AI/ML Intern

Jun 2025 – Aug 2025

Generative AI Stereo Synthesis Pipeline

- Engineered an end-to-end image generation pipeline using state-of-the-art diffusion models to accelerate the company's stereo camera R&D by providing highly domain-specific training datasets. Implemented strict conditioning controls to minimize hallucinations and preserve left-eye features for geometry- and pixel-accurate right-eye generation.
- Optimized generative architectures for 2K stereo synthesis under strict compute limits, and presented technical trade-off analyses to drive cross-functional product decisions.

University of California, Irvine | Research Assistant

Jul 2024 – Aug 2025

Biomedical Image Segmentation Architecture

- Developed a cascaded 3D U-Net pipeline in PyTorch, designing a sequential learning architecture where each stage leveraged prior predictions (liver → tumor → resection).
 - Co-authored an accepted paper for IEEE ISBI 2026 on benchmarking automated 3D segmentation models for liver surgical planning.
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SKILLS

Languages: Python, C++, C#, JavaScript, TypeScript

AI & Machine Learning: PyTorch, LangGraph, ChromaDB, Ragas, Cohere, SAM2, LLMs (OpenAI, Gemini)

Web & Cloud: React.js, AWS (ECS)

Developer Tools: Git, Docker, Linux/Unix